Reference trajectories

Augmented Predictive and Sliding-Mode Control for Cooperative Trajectory Tracking of Underactuated USV-AUV Systems in Multistage Missions

The USV follows a counterclockwise circle with periodically varying tangential speed:

$$\boldsymbol{p}_{d,u}(t) = R\begin{bmatrix}\cos\vartheta_d(t)\\ 1+\sin\vartheta_d(t)\end{bmatrix},$$
$$\dot{\vartheta}_d(t) = \frac{2\pi}{300}\left[1+0.454\sin\left(\frac{\pi t}{25}\right)-0.09\cos\left(\frac{2\pi t}{25}\right)\right],$$
$$R = 26.26\ \mathrm{m}, \qquad \vartheta_d(0) = -\frac{\pi}{2}.$$

The AUV horizontal reference is the transmitted USV position, with the position reference given by

$$\boldsymbol{p}_{d,a}(t_k) = \begin{bmatrix}x_u(t_k)\\ y_u(t_k)\\ z_d(t_k)\end{bmatrix}.$$

Its depth reference is

$$z_d(t) = \begin{cases}-\zeta(t), & 0 \le t < 100,\\ -18, & 100 \le t < 200,\\ -18+\zeta(t-200), & 200 \le t \le 300.\end{cases}$$

where

$$\zeta(\tau) = 0.25\int_0^{\tau} c\left(\frac{s}{28}\right)c\left(\frac{100-s}{28}\right)\,\mathrm{d}s,$$
$$c(r) = \begin{cases}0, & r < 0,\\ \dfrac{1-\cos(\pi r)}{2}, & 0 \le r \le 1,\\ 1, & r > 1.\end{cases}$$

The search, operation, and return stages occupy 0-100 s, 100-200 s, and 200-300 s, respectively. The prescribed working depth is 18 m below the surface.

Environmental disturbances

For the USV,

$$\boldsymbol{d}_u(t) = \begin{bmatrix} 1.3 + 2\sin(0.02t) + 1.5\sin(0.1t) \\ 0.6\sin(0.3t + \pi/6) \\ -\sin(0.09t + \pi/3) - 4\sin(0.01t) \end{bmatrix}.$$

For the AUV,

$$\boldsymbol{d}_a(t) = 0.05 \begin{bmatrix} m_{a11} d(t) \\ m_{a22} d(t) \\ m_{a33} d(t) \\ m_{a55} d(t) \\ m_{a66} d(t) \end{bmatrix},$$

$$d(t) = \sin(0.02t + \pi/3).$$

The coefficients m_{a11}, m_{a22}, m_{a33}, m_{a55}, and m_{a66} denote the corresponding diagonal entries of the AUV inertia matrix.

Vehicle	Disturbance component order
USV	Surge force, sway force, yaw moment
AUV	Surge force, sway force, heave force, pitch moment, yaw moment

Time is in seconds, positions are in metres, force components are in N, and moment components are in N m.